Davronbek Malikov

LLM Engineer | RAG Engineer | Agentic AI Engineer | ML Engineer | Ph.D.
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Summary

LLM, RAG, and Agentic AI Engineer with a Ph.D. in AI Convergence Engineering and 5 first-author publications. Leading the AI engineering team at TripleH, shipping on-premise multi-agent systems for enterprise clients. I build LLM systems end to end — post-training (SFT, DPO, RLHF), hybrid RAG, LangGraph agents, evaluation and observability (LangSmith, Arize Phoenix) — and deploy them on cloud and on-premise infrastructure.

Skills

- **LLM & Post-Training:** SFT, LoRA/QLoRA, DPO, RLHF (reward modeling, PPO, RLOO); quantization (AWQ, GPTQ, GGUF); serving (vLLM, TGI, Ollama, llama.cpp); open-weight models (Llama 4, Qwen3) and Claude/OpenAI APIs; structured outputs, function calling, prompt caching.
- **RAG:** hybrid search (dense + BM25, Reciprocal Rank Fusion), cross-encoder reranking, hierarchical and parent-child chunking, GraphRAG, permission-aware retrieval, cited answers; pgvector, Qdrant, Milvus.
- **Agentic AI:** LangGraph, LangChain; Supervisor/Router/Network multi-agent patterns, human-in-the-loop checkpoints, Judge agents, MCP, A2A.
- **Evaluation & Observability:** LangSmith (tracing, datasets, experiments), Arize Phoenix, OpenTelemetry and Jaeger, Ragas, DeepEval, LLM-as-a-Judge, golden test sets, token-cost and latency tracking; guardrails and prompt-injection defense.
- **Engineering:** Python, TypeScript, PyTorch, Hugging Face (Transformers, PEFT), FastAPI (async, SSE), PostgreSQL, Redis, Kafka, Docker, GitHub Actions, Next.js, Vercel.
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, SHAP, Optuna, clustering, Computer Vision (YOLO, OpenCV).

Experience

AI Engineer & Team Lead — TripleH, Seoul Dec 2025–Present

- Lead the AI engineering team building production autonomous agent systems for enterprise clients, from research and prototyping to on-premise and cloud deployment; set engineering standards, run code reviews, and mentor engineers.
- **Multi-Agent Orchestration Platform:** central hub that plans, routes, and monitors specialized sub-agents (automated market reports, a financial-document RAG assistant, an enterprise-messenger chatbot).
- **Legal Research Assistant:** multi-LLM ensemble that queries several models in parallel, aggregates their answers, and verifies every cited regional legal precedent before it reaches the user.
- **On-Premise AI Platform:** every agent runs locally on open-weight LLMs (Llama 4, Qwen3) with no external API calls — zero data leakage; also delivered a meeting-intelligence agent (on-premise speech-to-text, agenda-based minutes) and per-employee AI assistants.
- **Stack:** SFT/LoRA and DPO, hybrid-search RAG (pgvector, Qdrant), LangGraph and LangChain, monitoring with LangSmith and Arize Phoenix.

Independent Researcher (Sports Analytics) & AI Educator — Self-directed and online 2020–2025

- Researched ML and LLMs for soccer analytics (performance, salary, tactics); taught ML, deep learning, and LLM courses and mentored students on end-to-end AI projects.

Graduate Researcher — Gyeongsang National University, South Korea 2018–2025

- Big Data Systems Software Lab (2022–2025): ML for soccer analytics — salary, performance, and injury prediction. Distributed Systems Lab (2018–2021): mathematics for distributed systems. 5 first-author papers (see Publications).

Mathematics Instructor — Fergana, Uzbekistan 2014–2017

- Prepared students for university entrance exams, the SAT, and international mathematics exams.

Selected Projects

ERA Academy — AI EdTech platform, multi-agent engine, permission-aware RAG [GitHub] · [**Live**]

- One provider-agnostic agent engine behind three role-scoped assistants (tutor, teacher, admin): Claude, OpenAI, and local Ollama adapters, SSE streaming, usage tracking, and cost and rate limits.
- RAG: validated PDF/DOCX/Markdown ingestion → heading-aware chunking → embeddings → pgvector; access rules enforced inside the retrieval SQL query, cited answers, prompt-injection defense.
- 206 automated tests (authorization, HTTP security, AI engine, RAG), all passing. Next.js 15, Prisma, PostgreSQL.

Touchline — AI scouting copilot (Claude, LangGraph, hybrid RAG, LangSmith) [GitHub]

- Claude agent on LangGraph over 7 schema-validated tools (the model never writes SQL), with human-in-the-loop shortlist approval that resumes from a persisted checkpoint.

- Hybrid RAG (dense + BM25, RRF) in Qdrant with Claude citations; DuckDB warehouse of 50,149 players joined with StatsBomb event data (18 per-90 metrics for 594 players).
- Every run traced in LangSmith; 120-question golden set run as LangSmith experiments (filter accuracy, precision/recall@10, numeric grounding, invented-player detection); 102 automated tests.

TeachHelp — Post-training an LLM into an Uzbek–English teaching assistant [GitHub] · **[Live]**

- Full post-training of Qwen2.5 (SFT → DPO → RLHF), 14 models compared on one 68-question held-out test; SFT raised Qwen2.5-1.5B from 15/68 to 44/68.
- DPO from scratch: diagnosed v1's length bias (chosen shorter in 63% of pairs) and fixed it in v2 with length-balanced pairs and an SFT loss term — 44/68, all harmful and cheating requests refused.
- RLHF with Bradley–Terry reward models and PPO/RLOO across 4 judges, with reward-hacking checks; deployed as an English–Uzbek chat on Vercel with a Kaggle GPU backend.

Portiq — Multi-agent job hub (LangGraph, Kafka, A2A, pgvector) **[Live]**

- LangGraph orchestrator routes tasks to 11 agents over Kafka and A2A; a Judge agent sends weak output back for retry.
- Job pipeline (Saramin, hh.uz, Telegram) → normalisation and scam filter → pgvector match scoring; mock interviews in 4 languages; OpenTelemetry and Jaeger tracing across agents.

Soccer Salary Predictor & TacticalDNA — supervised and unsupervised ML [GitHub] · [GitHub]

- Leakage-safe salary pipeline with a stacking ensemble, Optuna tuning, and SHAP explanations (17% accuracy gain, SCIE 2025). TacticalDNA clusters 6,314 player-seasons; a blind stat line matched prime Messi at 98% cosine similarity.

More on the portfolio: TACTIQ (evidence-first football RAG, in development), and RAG-powered apps Palov, Hamyon, and Kotib — davronbek-portfolio.vercel.app

Publications (first author)

- *Predicting Soccer Player Salaries with Both Traditional and Automated Machine Learning Approaches.* Applied Sciences (SCIE), 2025. [DOI]
- *Beyond xG: A Dual Prediction Model for Analyzing Player Performance Through Expected and Actual Goals in European Soccer Leagues.* Applied Sciences (SCIE), 2024. [DOI]
- *Comparative Evaluation of Machine Learning Models for Predicting Soccer Injury Types.* KSCIE, 2024. [DOI]
- *The Application of Machine Learning on the Injury Prediction of Soccer Players.* ISE@APSEC, 2022. [PDF]
- *On the Construction and Properties of Lattice-Group Structure in Cartesian Product Spaces.* Journal of Computer Science, 2020. [DOI]

Education

Ph.D. in AI Convergence Engineering, Gyeongsang National University, Republic of Korea 2020–2025
Thesis: *Advancements in Soccer Analytics: Players' Goals and Salary Prediction Models with Machine Learning*
Master of Informatics, Gyeongsang National University, Republic of Korea 2018–2020
Bachelor of Mathematics, Fergana State University, Uzbekistan 2014–2018

Languages

English (Fluent), Uzbek (Native), Turkish (Advanced), Russian (Intermediate), Korean (Conversational)